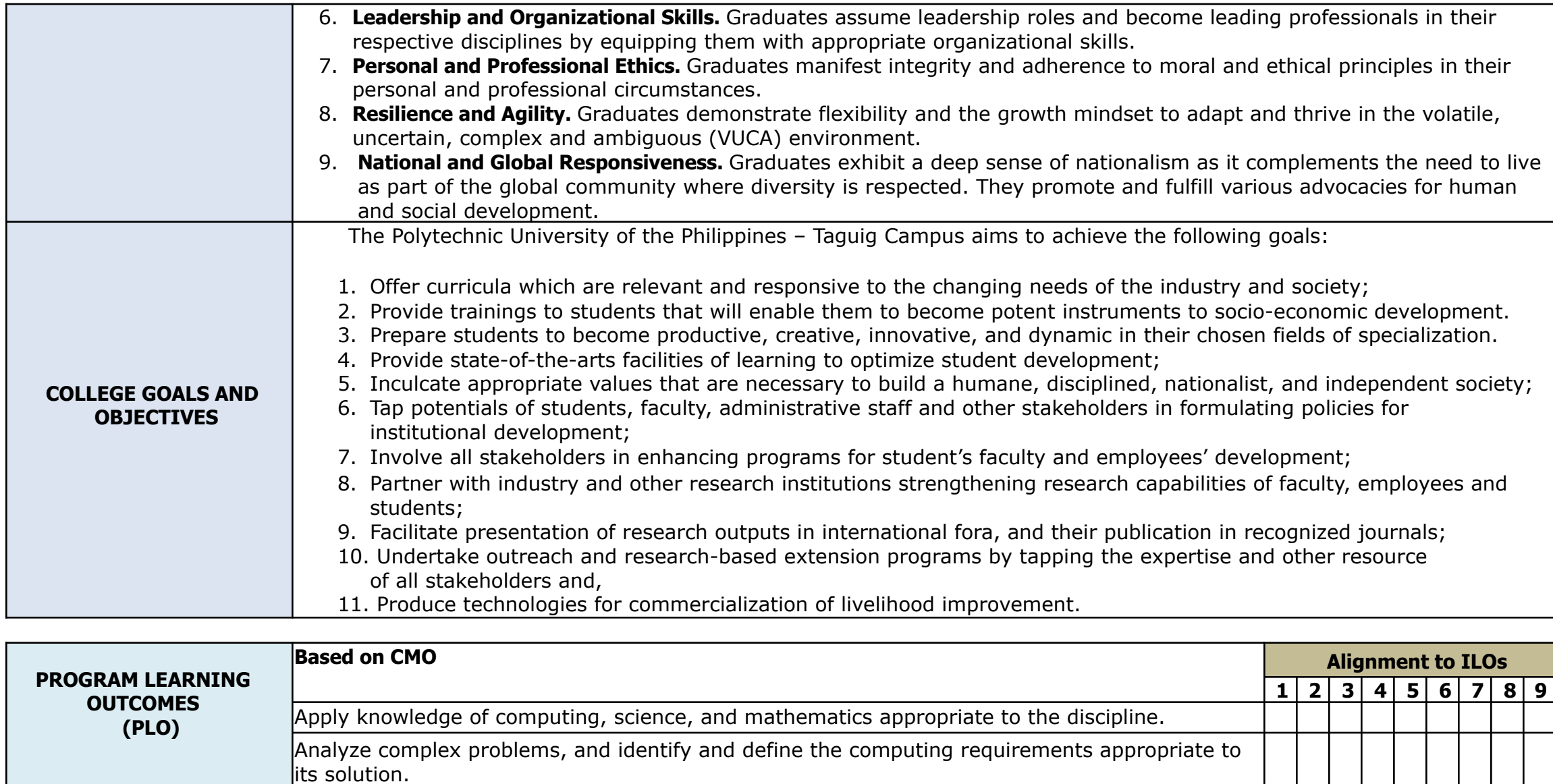
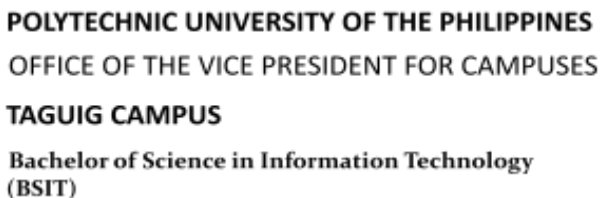




COURSE SYLLABUS						
Cours Code	COMP 009	Course Title	Object-Oriented Programming		Course Credit	3
Course Description		This course provides a comprehensive study of Object-Oriented Programming (OOP) principles and methodologies. It emphasizes the design and implementation of modular, reusable, and maintainable software using objects and classes. Core topics include encapsulation, inheritance, polymorphism, and abstraction, with practical applications developed in a high-level programming language such as Java, C++, or Python.				
Pre-Requisites		COMP 003	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



INSTITUTIONAL LEARNING OUTCOMES (ILO)	As a polytechnic state university, PUP shall develop its students to possess: 1. Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. 2. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. 3. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors. 4. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. 5. Passion for Lifelong Learning. Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.
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	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									



	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)	Explain the fundamental concepts of Object-Oriented Programming, including classes, objects, inheritance, polymorphism, abstraction, and encapsulation.	Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Design object-oriented solutions to real-world problems using appropriate modeling techniques such as UML class and sequence diagrams.								
	Implement well-structured programs using OOP principles in a high-level programming language (Java).								
	Apply inheritance and polymorphism to promote code reuse, flexibility, and maintainability in program design.								
	Demonstrate the use of constructors, methods, interfaces, and exception handling to build robust and efficient applications.								



	Utilize file handling, data structures, and collections within OOP frameworks to manage and process data effectively.							
	Evaluate the performance and quality of object-oriented programs based on readability, reusability, and scalability.							
	Collaborate effectively in teams to plan, develop, test, and document object-oriented software projects.							



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP. - Gain an overview of the course subject and requirements.	- Orientation about PUP Vision, Mission, and 10 Pillars. - Overview about the Subject Course and Course Requirements.	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X	X		
2	- Apply basic syntax and structure of an OOP language (Java/C++/Python) - Understand OOP fundamentals and importance	- IDE setup, writing first program, data types, operators. - Introduction to Programming Paradigms OOP Concepts.	Hands-on coding in lab	Tutorial videos, coding practice	Lab exercise	Coding submission	X	X		
3	Use decision-making and looping constructs in problem-solving	Control structures (if, else, switch, loops).	Guided coding problems	Online coding challenges	In-class coding quiz	Problem set submission	X	X		



4	• Apply functions/methods to modularize code	• Functions, parameter passing, return types.	Lecture + exercises	Recorded demos, practice problems	Seatwork (coding)	Online practice quiz		X		
5	• Demonstrate understanding of classes and objects	• Defining classes, instantiating objects.	Lecture + hands-on demo	Online discussion, coding activity	Quiz + coding lab	Code submission		X	X	
6	• Implement constructors and destructors	• Constructor overloading, destructors.	Coding demo	Self-paced coding tasks	Lab activity	Coding assignment		X	X	
7	• Apply encapsulation using access modifiers	• Access specifiers, getters & setters.	Lecture + guided coding	Case-based activity	Quiz	Online coding task		X	X	



8	- Demonstrate inheritance in real-world scenarios - Implement polymorphism	- Single & multiple inheritance, super keyword. - Method overloading, overriding, and dynamic binding.	Lecture + exercises	Reading material + coding tasks	Quiz + coding activity	Assignment submission		X	X	
9	MIDTERM WRITTEN EXAMINATION									
10	Apply abstraction in program design	Abstract classes, interfaces	Lecture + case studies	Self-study coding tasks	Case analysis	Coding activity		X	X	
11	Work with arrays and collections	Arrays, ArrayList, Vectors	Lab coding demo	Practice problems online	Lab activity	Online quiz		X	X	
12	Apply exception handling	Try-catch-finally, custom exceptions	Guided coding	Recorded lecture, exercises	Coding exercise	Online coding submission		X	X	
13	Demonstrate file handling	File I/O, serialization	Lab exercise	Practice activities	Hands-on activity	File-handling assignment		X	X	
14	Work with advanced OOP features	Packages, static members	Lecture + demo	Self-study reading & coding	Short quiz	Coding practice			X	
15	Apply GUI basics in OOP	Introduction to GUI programming	Coding lab	Online practice	Lab activity	GUI coding task			X	
16	Plan and design an OOP-based project	Project guidelines, design principles	Lecture + consultation	Self-directed project planning	Project proposal	Online submission				X
17	Implement and	Debugging,	Lab consultations	Online	Project defense	Project				X



	test OOP project • Present and evaluate project outputs	testing • Project presentation & documentation	Face-to-face presentation	mentoring Online submission of docs	rehearsal Final defense	progress report Project documentation				
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18

FINAL WRITTEN EXAMINATION

LIST OF REFERENCES

BOOKS:

WEBSITE:

W3Schools.com. (n.d.). https://www.w3schools.com/java/java_oop.asp

Object-Oriented Programming concepts. (2022, February 3). Coursera. <https://www.coursera.org/learn/concepts-of-object-oriented-programming>

Amos, D. (2024, December 15). Object-Oriented Programming (OOP) in Python. <https://realpython.com/python3-object-oriented-programming/>

Object-Oriented Programming ONRAMP | Self-Paced Online Courses - MATLAB & Simulink. (n.d.). <https://matlabacademy.mathworks.com/details/object-oriented-programming-onramp/oroop>

edX. (n.d.). Best Online Object-Oriented Programming Courses & Programs | edX. <https://www.edx.org/learn/object-oriented-programming>

Nathaniel Stickman. (2022, June 10). *A Guide to OOP Principles in Java*. Linode Docs. <https://www.linode.com/docs/guides/oop-principles/>



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will



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	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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**2025**
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

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Rank 21
GENERATIVE AI APPLICATION
Rank 39
STUDENT SUPPORT ENGAGEMENT
Rank 50
SYMBOLIC DISCUSSION

**QS STARS**



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- Social Responsibility
- Online Learning
- Academic Development
- Special Criteria Bachelor of Science in Nutrition and Dietetics (BSND)
- Internationalization

**5★**
Times Higher Education Impact Rankings 2024
DEBUTS AT 1501+



	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
Students must have completed an introductory programming course and possess basic knowledge of algorithms and control structures. They are required to use a computer with a programming IDE, such as Eclipse or Visual Studio Code, and perform coding activities, projects, and assessments using a high-level programming language, such as Java, C++, or Python.	Mid Term	Final Term
	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper Midterm Examinations 30% 100%	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper Final Examinations (Teaching Demonstration) 30% 100%
	Final Grade = (Midterm Grade + Final Term Grade) / 2	
CLASS INFORMATION		FACULTY INFORMATION
Course, Year, and Section: BSIT-TG 2-1		Name of Faculty: FRANCO, FRANCIS FRADES
Day and Time: F/F 07:30 AM - 10:30 AM / 10:30 AM - 12:30 PM		Consultation Time:
Room: TGA205		Office Tel. No./ Mobile Phone No.
School Year: 2025-2026 Semester: Second		E-Mail Address:



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Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
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